HRV Analysis API 1.0.0 OAS 3.1

/openapi.json

Calculate HRV RMSSD from PPG signal data with quality checks

default

GET / Root

Root endpoint with API information

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

GET `/health` Health Check

Health check endpoint

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema `"string"`	*No links*

POST `/analyze` Analyze Hrv

Analyze PPG data and calculate HRV RMSSD with signal quality assessment.

This endpoint accepts PPG signal data and calculates the RMSSD (Root Mean Square of Successive Differences) metric for Heart Rate Variability analysis. It performs signal quality checks by dividing the data into 3-second segments and rejecting windows with too many bad segments.

Args: request: HRVRequest containing ppg_data, sampling_rate, and max_bad_segments

Returns: HRVResponse with rmssd value and bad_segments count on success

Raises: HTTPException (422): If signal quality is too poor or HRV calculation fails

Parameters Try it out

No parameters

Request body required

application/json

Example Value Schema

```
{
  "ppg_data": [
    438,
    445,
    452,
    460,
    468
  ],
  "sampling_rate": 100,
  "max_bad_segments": 0
}
```

Responses

Code	Description	Links
200	Successful HRV calculation Media type application/json Controls `Accept` header. **Example Value** Schema `{ "success": true, "rmssd": 0, "bad_segments": 0 }`	*No links*
422	Signal quality too poor or calculation failed Media type application/json **Example Value** Schema `{ "success": false, "error": "string", "bad_segments": 0 }`	*No links*

Schemas

HRVErrorResponse ^ Collapse all `object`

Response model for failed HRV calculation

`success` ^ Collapse all `boolean`

Indicates calculation failure

Default=false

`error*` ^ Collapse all `string`

Error message describing what went wrong

`bad_segments*` ^ Collapse all `integer`

Number of bad 3-second segments detected

HRVRequest ^ Collapse all `object`

Request model for HRV analysis

`ppg_data*` ^ Collapse all `array<number>` **≥ 1000 items**

Array of PPG signal values

`Items` `number`

Example > Expand all `array`

`sampling_rate*` ^ Collapse all `number` **(0, 1000]**

Sampling rate in Hz

Example=*100*

`max_bad_segments` ^ Collapse all `integer` **≥ 0**

Maximum number of bad 3-second segments allowed (default: 0)

Default=0

Example=*0*

HRVResponse ^ Collapse all `object`

Response model for successful HRV calculation

`success` ^ Collapse all `boolean`

Indicates successful calculation

Default=true

`rmssd*` ^ Collapse all `number`

Calculated RMSSD value in milliseconds

`bad_segments*` ^ Collapse all `integer`

Number of bad 3-second segments detected